A numerical analysis method based on flow continuity for hydrodynamic journal bearings

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A numerical method of hydrodynamic bearing analysis is presented which is simple in concept, yet capable of development to handle complex situations such as dynamic misalignment. It is similar to the finite difference solution of Reynolds equation, but incorporates a more realistic modelling of cavitation. The approach to a numerical solution is direct, and should facilitate a better 'feel' for the way in which the physical processes are modelled. Results produced with this analysis are compared with other published data for aligned crankshaft bearings and misaligned sterntube bearings.

Keywords: *journal bearings, numerical analysis, hydrodynamic bearings, cavitation*

Hydrodynamic journal bearing operation, when considered in detail, is complex when factors such as oil film viscosity variation with temperature, pressure and shear rate, bearing surface elasticity, differential thermal expansion, etc., are taken into account. The choice of which factors should be included and which neglected should reflect the operating conditions of the bearing under consideration. In a sterntube bearing, for example, the low rotational speeds, moderate loading and effective cooling of the surrounding structure by sea water make the assumption of an isoviscous oil film reasonable. This results in a considerable simplification of the analysis in relation to any variable viscosity model.

For crankshaft bearing analysis, the assumption of isoviscosity is more open to question. Since the effects of temperature and pressure on oil film viscosity act in partial opposition, the author considers an isoviscous oil film to be an acceptable model for moderately loaded crankshaft bearings.[1] In practice the complexity of any analysis is limited by computing power requirements.

For moderate speeds where laminar flow conditions may be assumed and lubricant inertia effects neglected, many authors have used the finite difference solution of Reynolds equation[2-4]. When a numerical solution is adopted, however, it is not necessary to start with Reynolds equation. In this paper a direct approach to the numerical solution is made which is considered simpler and physically more meaningful. Reynolds equation does not in fact take cavitation into account, and some authors,[2,4] have simply truncated the predicted sub-cavitation pressure region. Depending upon the oil groove geometry and journal location, this model may lead to substantial inaccuracy in the predicted position of the downstream cavitation boundary. This is due to the implicit assumption that a full oil film is available to generate hydrodynamic pressure as soon as film convergence commences. Such an assumption may be at considerable variance with continuity requirements.

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In the solution presented in this paper, the oil film is divided into rectangular components referred to as elements. A continuity equation is written for each element in terms of the pressure at its centre and that of the surrounding four elements or boundaries. Linear pressure gradients are assumed between the element centres, or between the centre and an adjacent film boundary. This is satisfactory, provided the element dimensions are sufficiently small in relation to the rate of change of pressure gradient.

Within the cavitation zone a constant cavitation pressure (usually assumed to be atmospheric) is specified, and a gas/vapour flow term is introduced to satisfy continuity in this area. Film pressure distribution derived by this method approximates to the Swift–Steiber condition of zero pressure gradient at the cavitation zone boundaries. The above condition is not satisfied exactly because the circumferential positions of the cavitation zone boundaries are located only to an accuracy of $\pm \Delta c$. Specification of the zero pressure gradient condition is, however, unnecessary with this method. The only boundary conditions which need be defined are bearing ambient pressure, supply groove pressure and location and cavitation pressure.

The film pressure matrix is determined by a Gauss–Seidel iterative solution of the above equations using successive over relaxation. Derivation of the film pressure matrix is the fundamental part of the analysis, and computationally the most time consuming. Calculation of the total oil film force and moment components, flow rate, power loss and displacement and velocity coefficients is relatively straight forward.

Description of the analysis

Assumptions

For the analysis presented here, the following assumptions have been made:

- Laminar flow.
- Newtonian lubricant with constant effective viscosity within the oil film.
- Rigid and perfectly circular journal and bearing surfaces.

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- Pressure gradients between adjacent oil film element centres, or between the element centre and a specified adjacent film boundary, are assumed to be linear. These linear pressure gradients are assumed to be the effective values for the element boundary they cross for the purpose of calculating pressure-induced flow rate.
- Oil film rupture occurs at a specified cavitation pressure which remains constant throughout the ruptured region.
- Gas and/or vapour cavities may form or collapse instantaneously in order to satisfy continuity requirements and the specified cavitation pressure.

Film geometry equations

For any axial position in a journal bearing, the film thickness at angle θ from the bearing top may be calculated from:

$$h = C_d/2 + e\cos(\theta - \psi) \quad (1)$$

Where the bearing is subject to misalignment, e and ψ are functions of axial position and may be determined via the components of eccentricity in the vertical and horizontal planes thus:

$$e_s = [e_{sy}^2 + e_{sx}^2]^{1/2} \quad (2)$$

Notation

A_{xx}, etc.	Displacement (stiffness) coefficient, N/m
B_{xx}, etc.	Velocity (damping) coefficient, N s/m
B_{ai}, B_{ao}	Effective width of oil stream at inlet to and outlet from an element, allowing for presence of gas/vapour, m
C_d	Diametral clearance, m
D	Journal diameter, m
e	Journal eccentricity (general), m
e_{cy}, e_{cx}	Vertical and horizontal components of journal eccentricity at bearing axial centre, m
e_{sy}, e_{sx}	As above but at distance s from the bearing axial centre (positive to the left), m
F_c	Circumferential viscous shear force acting on an element, N
F_y, F_x	Vertical and horizontal components of total oil film force acting on journal, N
h	Oil film thickness (general), m
h_a, h_b, h_c, h_d	Oil film thickness at each corner of the element. For element J, I these thicknesses correspond respectively to $(h(J,I), h(J,I+1), h(J+1,I), h(J+1,I+1)$, m
H	Total power absorbed by oil film, N m/s
H_{ci}, H_{co}	Element inlet and outlet circumferential pressure flow functions, m^5/N s
H_{ai}, H_{ao}	As above but for axial flow, m^5/N s
I	Axial element position reference
J	Circumferential element position reference
K	Nett velocity-induced flow out of the element, m^3/s
L	Bearing length, m
M_c	No of element rows (circumferential positions)
M_y, M_x	Vertical and horizontal components of total oil film moment acting about the bearing axial centre. The terms vertical and horizontal refer to the planes in which the moments act, N m
N	Journal rotational speed, r/s
N_a	No of element columns (axial positions)
P	Film pressure at the element centre, N/m^2
P_c	Cavitation pressure, N/m^2
P_p	Value of P for a given element during the previous iteration, N/m^2
P_s	Supply pressure, N/m^2
$P_{ci}, P_{ai}, P_{ao}, P_{ao}$	Film pressures at the centres of the elements adjacent to the element under consideration. For element J, I these correspond respectively to $P(J-1,I)$, $P(J,I-1), P(J+1,I), P(J,I+1)$, N/m^2
Q_{vi}, Q_{vo}	Inlet and outlet element gas/vapour flow rates, m^3/s
Q_{ci}, Q_{co}	Inlet and outlet element circumferential lubricant flow rates (assuming a full lubricant film), m^3/s
Q_{ai}, Q_{ao}	Inlet and outlet element axial lubricant flow rates, m^3/s
s	Axial distance from bearing axial centre, m
U	Journal surface velocity due to rotation, m/s
V_N, V_T	Components of journal axis velocity normal and tangential to the element, m/s
$V_y(I), V_x(I)$	Vertical and horizontal components of journal axis velocity at the axial position corresponding to centre of element column I, m/s
$\bar{W}, \bar{H}, \bar{Q}_s$	Dimensionless load, power loss and side leakage flow (Fig 4)
y, x	Vertical and horizontal components of journal lateral displacement, m
$\dot{y}, \dot{x}$	Vertical and horizontal components of journal lateral velocity, m/s
α_y, α_x	Vertical and horizontal components of journal angular misalignment, rad
γ, λ	Angular displacement of journal axis in vertical and horizontal planes, rad
$\dot{\gamma}, \dot{\lambda}$	Angular velocity of journal axis in vertical and horizontal planes, rad/s
$\Delta a, \Delta c$	Element lengths in the axial and circumferential directions, m
ϵ	Eccentricity ratio $2e/D$
η	Dynamic viscosity, $N s/m^2$
θ	Angular distance from bearing top to required circumferential position, degrees
τ	Mean circumferential viscous shear stress on element journal surface, N/m^2
ψ	Attitude angle, degrees
ω	Journal rotational speed, rad/s

The main geometrical parameters are illustrated in Figs 1(a), (b) and (c).

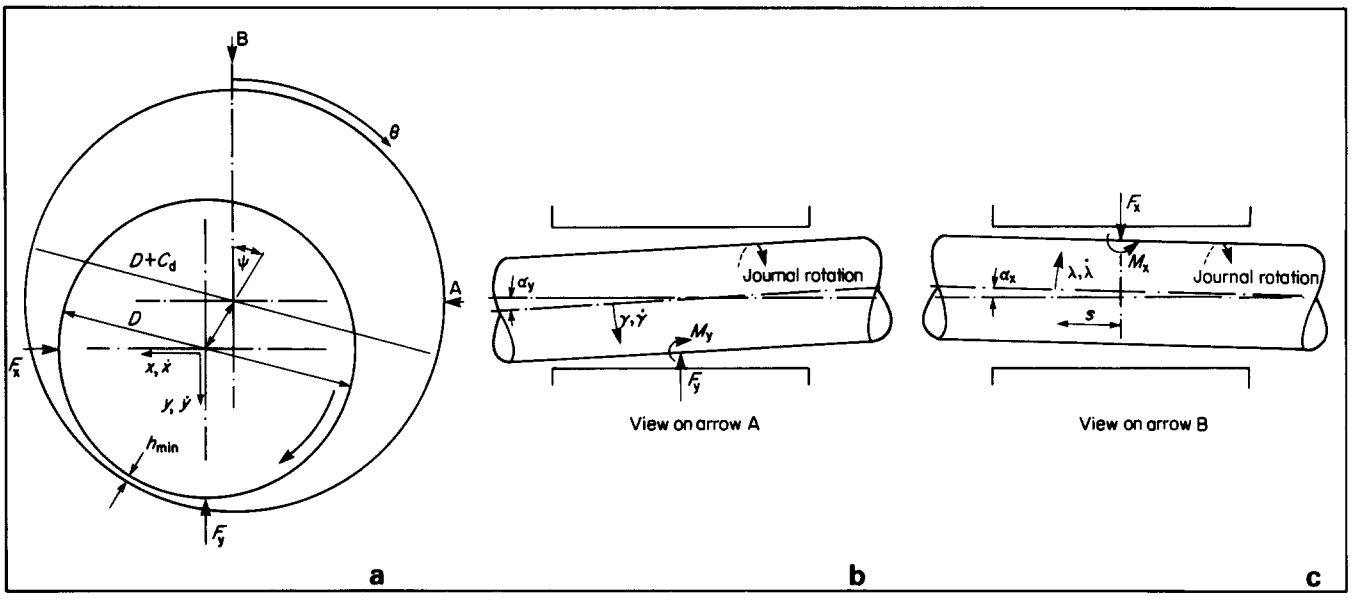


Fig 1 Sign convention for bearing: (a) end view; (b) view on vertical plane; (c) view on horizontal plane

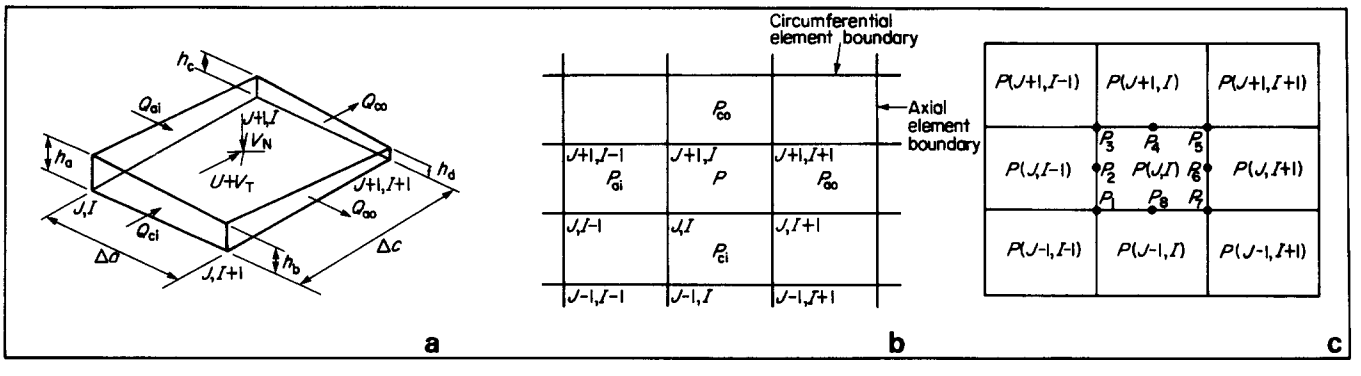


Fig 2 Oil film element details

$$\psi_s = \tan^{-1}(e_{sx}/e_{sy}) \tag{3}$$

where

$$e_{sy} = e_{cy} + s\alpha_y \tag{4}$$

$$e_{sx} = e_{cx} + s\alpha_x \tag{5}$$

For a specified journal eccentricity and misalignment, the film thickness h may thus be determined at any circumferential and axial position.

Element continuity equation

In Fig 2(a) an element of the oil film is shown, and in Fig 2(b) its circumferential and axial position reference system in relation to the neighbouring elements is specified. A continuity equation may be written for each element thus:

$$-H_{ci}(P - P_{ci}) - H_{ai}(P - P_{ai}) - Q_{vi} = K - H_{co}(P_{co} - P) - H_{ao}(P_{ao} - P) - Q_{vo} \tag{6}$$

Derivation of Eq (6) is given in Appendix 1. Outside the cavitation zone, $P > P_c$ and $Q_{vo} = 0$, therefore Eq (6) may be reduced to the following form:

$$P = \frac{H_{ai}P_{ai} + H_{ci}P_{ci} + H_{ao}P_{ao} + H_{co}P_{co} - K - Q_{vi}}{H_{ai} + H_{ci} + H_{ao} + H_{co}} \tag{7}$$

where

$$H_{ci} = \frac{(h_a + h_b)^3}{96\eta} \cdot \frac{\Delta a}{\Delta c} \tag{8}$$

$$H_{ai} = \frac{(h_a + h_c)^3}{96\eta} \cdot \frac{\Delta c}{\Delta a} \tag{9}$$

$$H_{co} = \frac{(h_c + h_d)^3}{96\eta} \cdot \frac{\Delta a}{\Delta c} \tag{10}$$

$$H_{ao} = \frac{(h_b + h_d)^3}{96\eta} \cdot \frac{\Delta c}{\Delta a} \tag{11}$$

$$K = (h_c + h_d - h_a - h_b) \cdot (U + V_T) \cdot \frac{\Delta a}{4} - V_N \cdot \Delta a \cdot \Delta c \tag{12}$$

In Eq (12) the terms V_T and V_N are the components of journal axis lateral velocity tangential and normal to the surface at the element centre. These may be calculated from:

$$V_N = -[V_y(I)\cos\theta + V_x(I)\sin\theta] \tag{13}$$

$$V_T = V_y(I)\sin\theta + V_x(I)\cos\theta \tag{14}$$

The journal axis lateral velocity is expressed as vertical and horizontal components which are a function of axial position for a journal subject to angular velocity of its axis:

$$V_y(I) = \dot{\gamma}\,\frac{L}{2}\,[1 - \frac{2}{N_a}(I - 0.5)] + \dot{y} \tag{15}$$

$$V_x(I) = \dot{\lambda}\,\frac{L}{2}\,[1 - \frac{2}{N_a}(I - 0.5)] + \dot{x} \tag{16}$$

Where an element is adjacent to a specified oil film boundary, (eg oil supply groove, bearing ends) the appropriate adjacent element pressure in Eq (7) is replaced by the specified boundary pressure. The corresponding pressure flow function value must then be multiplied by 2 since the pressure gradient is determined over half the relevant element dimension.

A clearer understanding of the above analysis may be gained by considering the following points:

- H_{ci} and H_{co} are functions which yield the upstream and downstream circumferential element pressure-induced flow rates respectively, when multiplied by the appropriate differential pressures. Strictly speaking the terms upstream and downstream in this context are arbitrary, and where reversal of the flow is encountered, the sign of the computed flow automatically changes.
- H_{ai} and H_{ao} are similarly related to the axial pressure-induced flow rates.
- K represents the nett circumferential flow out of the element induced by the velocity of the journal surface. As shown in Fig 2(a), the velocity of the journal surface comprises U due to the journal rotational speed and V_T due to the lateral velocity of the journal in direction tangential to the element centre, and V_N is the component of journal lateral velocity normal to the element centre.

Cavitation model

Where cavitation is present, the lubricant flow rates across the circumferential boundaries of an element may be reduced by the gas/vapour flow rate terms Q_{vi}, Q_{vo}. In these conditions the nett lubricant flow rates across the upstream and downstream boundaries are thus $Q_{ci} - Q_{vi}$ and $Q_{co} - Q_{vo}$, respectively. Derivation of Q_{vi} and Q_{vo} to satisfy continuity within the cavitation zone is as follows:

For any element, when Eq (7) predicts a pressure below the specified cavitation pressure (ie $P < P_c$), the predicted pressure is ignored and the element centre pressure is made equal to P_c. It should be noted that Eq (7) is based on the assumption that Q_{vo} is zero and that $P \geqslant P_c$, ie a full lubricant film at the element circumferential downstream boundary is assumed. Where substitution of P_c for P becomes necessary the above assumptions for Eq (7) are invalidated. Since this situation clearly indicates the presence of cavitation, it is postulated that continuity may be satisfied with $P = P_c$ by determining a finite value for Q_{vo}. For the circumferentially adjacent element in the downstream direction, the value of Q_{vi} will thus be equal to the above Q_{vo}, ie $Q_{vi}(J+1, I) = Q_{vo}(J, I)$. In general, when applying Eq (7), Q_{vi} is assumed to be zero unless a finite Q_{vo} has been calculated for the upstream circumferentially adjacent element. Where the $P = P_c$ substitution is made the value of Q_{vo} required to maintain continuity is calculated from:

$$Q_{vo} = K + Q_{vi} - (H_{ci}P_{ci} + H_{ai}P_{ai} + H_{co}P_{co} + H_{ao}P_{ao}) + P_c(H_{ci} + H_{ai} + H_{co} + H_{ao}) \quad (17)$$

Solution technique

Solution of the film pressure matrix is achieved by application of Eq (7) to each element using the Gauss–Seidel relaxation process with successive over relaxation. The computation flow diagram for this solution is shown in Fig 3. Cavitation zone boundaries are automatically determined in this process. No detailed study of the optimum over relaxation factor (orf) has been carried out, but a value of 1.7 was found to give the minimum number of iterations in a limited range of tests.

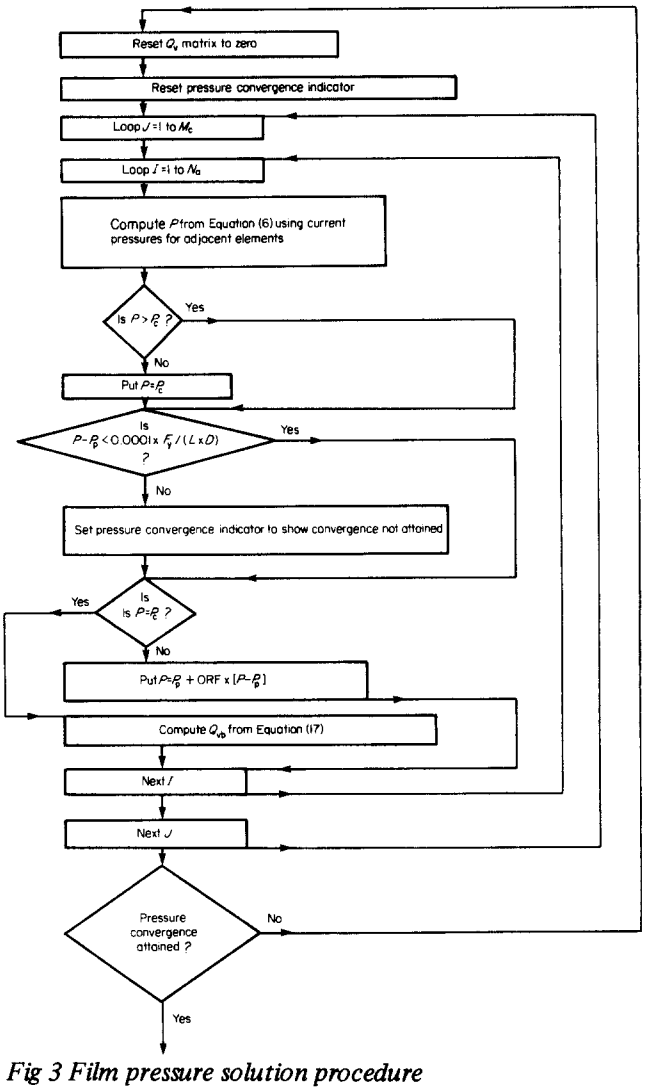


Fig 3 Film pressure solution procedure

Having determined the film pressure matrix, the total oil film force and moment vertical and horizontal components may be calculated by summation of the corresponding force and moment components for each element. To derive the oil film force on each element, the mean pressure is calculated from:

$$P_{mean} = \frac{4P + P_1 + P_2 + P_3 + P_4 + P_5 + P_6 + P_7 + P_8}{12} \quad (18)$$

where P_1 to P_8 are the pressures at each corner of the element and at the centre of each boundary, as shown in Fig 2(c). These pressures are calculated from the linear pressure gradients to the adjacent elements or oil film boundaries, ie using the J, I element reference system:

$$P_1 = [P(J, I) + P(J-1, I-1)]/2$$

$$P_2 = [P(J, I) + P(J, I-1)]/2 \quad \text{etc.}$$

Flow rate

Once the film pressure matrix has been determined, the flow rate across any element boundary may be readily calculated from:

$$Q_{ci} = (h_a + h_b)\,U\,\frac{\Delta a}{4} - H_{ci}(P - P_{ci}) \quad (19)$$

and

$$Q_{ai} = -H_{ai}(P - P_{ai}) \quad (20)$$

Similar equations for Q_{co} and Q_{ao} may be written, but with the exception of boundaries at $J = M_c$ and $I = N_a$, respectively, calculation of these values would simply duplicate the respective Q_{ci} and Q_{ai} values for the corresponding 'downstream' elements.

Power loss

Power loss is computed by summating the circumferential viscous shear force applied to the journal by each element. For this purpose, the circumferential shear stress on each element is calculated as the mean value of the shear stress at the circumferential boundaries. This results in:

$$F_c = U \Delta c \eta \left[\frac{B_{ai}}{h_a + h_b} + \frac{B_{ao}}{h_c + h_d} \right] + [(h_a + h_b)(P - P_{ci}) + (h_c + h_d)(P_{co} - P)] \Delta a/8 \quad (21)$$

$$H = U \sum_{J=1, I=1}^{J=M_c, I=N_a} F_c \quad (22)$$

where

$$B_{ai} = \Delta a - \frac{4Q_{vi}}{U(h_a + h_b)} \quad (23)$$

and

$$B_{ao} = \Delta a - \frac{4Q_{vo}}{U(h_c + h_d)} \quad (24)$$

B_{ai} and B_{ao} are the effective widths of the oil stream entering and leaving the element respectively, and will clearly be equal to the full width of the element Δa where no cavitation is present. For elements adjacent to a specified bearing oil film boundary, the appropriate pressures in Eq (21) are equated to the boundary pressure, and a factor of 2 is introduced to produce the correct pressure gradient. This process is similar to the corresponding modification of the film pressure Eq (7). It may be noted that in the right hand side of Eq (21), the expression:

$$U \Delta c \eta \left[\frac{B_{ai}}{h_a + h_b} + \frac{B_{ao}}{h_c + h_d} \right]$$

represents the shear force due to surface velocity-induced flow. The expression $[(h_a + h_b)(P - P_{ci}) + (h_c + h_d) \times (P_{co} + P)] \Delta a/8$ is the shear force due to pressure-induced flow, and clearly falls to zero in the cavitation zone.

Element grading

The above equations are for a constant element size. Smaller elements are desirable in areas where large changes in film pressure gradient occur. Straight forward modification of the equations may be introduced to vary the element dimensions in either the circumferential or axial directions. Grading of the element dimensions enables better modelling in areas of rapid change of film pressure gradient, without incurring a large increase in computing time. Lloyd *et al*[5] introduced circumferential grading in their finite difference solution of Reynolds equation by making the circumferential mesh length a function of film thickness. Where misalignment is present, circumferential grading becomes difficult due to variation in the circumferential position of minimum film thickness along the bearing axis. For heavy misalignment, axial grading is preferred, and the author has recently introduced this in his work on sterntube bearings. This has been achieved by making the element axial dimension increase as a linear function of the element column number (I) from the bearing left hand end to the centre, and in a similar symmetrical manner from the centre to the right hand end. A grading factor is defined as the ratio of axial element length for uniform size, to that of elements adjacent to the bearing ends, with grading and the same number of element columns.

Dynamic coefficients

To calculate oil film displacement and velocity coefficients, incremental displacements and velocities are applied to the journal. The film pressure relaxation and element force component summation subroutines are then repeated to determine the corresponding change in the oil film force and moment components. This is a quasi-static process and neglects the 'oil film history' effects outlined by Jones[6]. Since the displacement and velocity increments used are small, errors arising from the above procedure should be insignificant.

The terms displacement and velocity coefficient are alternatives to the commonly used 'stiffness' and 'damping coefficients'. Oil films exhibit a high degree of nonlinearity, use of 'stiffness' and 'damping coefficient' can therefore be misleading. In view of the nonlinearity, some workers[2,7] have adopted the terms displacement and velocity coefficient and have defined them with respect to zero amplitude. For computational purposes, finite amplitudes (incremental) must clearly be used, and must be large enough to produce a significant force or moment change in relation to the force and moment resolution accuracy of the numerical solution. To meet this requirement, and still obtain the best approximation to a 'zero' amplitude coefficient, the author uses equal positive and negative increments, and takes the mean value of the coefficients thus derived. This technique also facilitates computation of the coefficient gradients with respect to amplitude. These gradients are even more sensitive to force and moment resolution accuracy, and their practical usefulness is questionable.

Where dynamic misalignment conditions exist, 32 displacement and velocity coefficients are required:

$$\begin{bmatrix} F_y \\ F_x \\ M_y \\ M_x \end{bmatrix} = \begin{bmatrix} A_{yy} & A_{yx} & B_{yy} & B_{yx} & A_{y\gamma} & A_{y\lambda} & B_{y\gamma} & B_{y\lambda} \\ A_{xy} & A_{xx} & B_{xy} & B_{xx} & A_{x\gamma} & A_{x\lambda} & B_{x\gamma} & B_{x\lambda} \\ A_{\gamma y} & A_{\gamma x} & B_{\gamma y} & B_{\gamma x} & A_{\gamma\gamma} & A_{\gamma\lambda} & B_{\gamma\gamma} & B_{\gamma\lambda} \\ A_{\lambda y} & A_{\lambda x} & B_{\lambda y} & B_{\lambda x} & A_{\lambda\gamma} & A_{\lambda\lambda} & B_{\lambda\gamma} & B_{\lambda\lambda} \end{bmatrix} \times \begin{bmatrix} y \\ x \\ \dot{y} \\ \dot{x} \\ \gamma \\ \lambda \\ \dot{\gamma} \\ \dot{\lambda} \end{bmatrix} \quad (25)$$

Practical analysis program

In the above analysis technique it is necessary to specify journal eccentricity as input and to derive journal load as output data. The author's analysis iterates film viscosity to meet a specified load, and where a specified load direction is required (usually vertical), the attitude angle is iterated accordingly. This was found to be necessary since the performance is dependent upon the specified film boundary pressure, due to the influence of this parameter

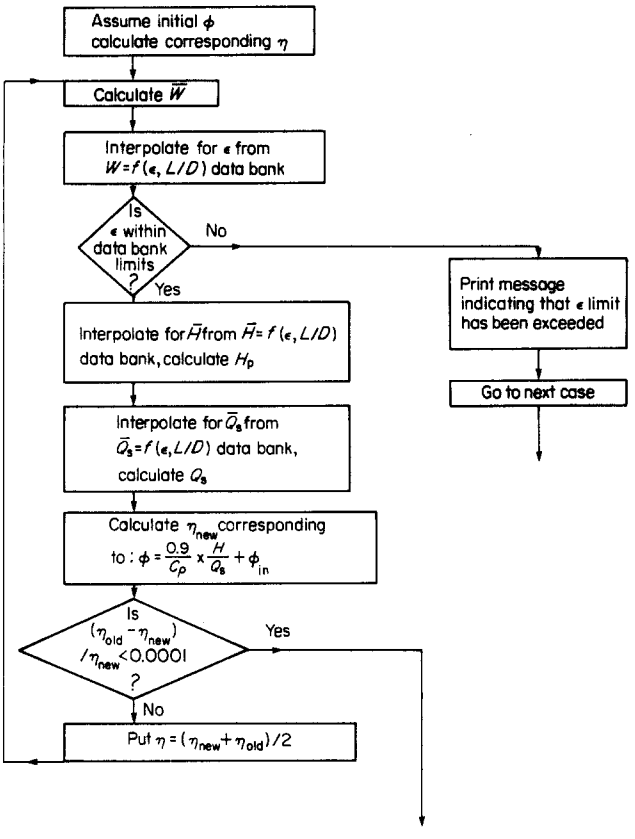


Fig 4 Practical analysis program. ϕ = lubricant temperature, $\bar{W} = (F_y/LD\eta N)(C_d/D)^2$, $\bar{H} = HC_d/\eta N^2 LD^3$, $\bar{Q}_s = Q_s L^2 \eta / C_d^3 F_y$, $\epsilon = 2e/C_d$, $C \cdot \rho$ = lubricant specific heat x density

on the extent of the cavitation zone. To generalize the data relating to this aspect of bearing performance, the boundary pressure was rendered dimensionless by division by the specific bearing pressure. The facility for specifying the journal load was therefore necessary to produce data for specified dimensionless boundary pressures.

Despite such refinements, the fact that journal eccentricity has to be specified as input data renders such a program unsuitable for practical analysis purposes. The main use of the above program is therefore to generate dimensionless performance data for building a data bank upon which a practical analysis program can be based. Such a program has been developed by the author for sterntube bearings and involves an iterative process in which the effective film viscosity is determined by the assumption that a fixed proportion of the heat generated is carried away by the oil. Operation of the program is thus essentially similar to the analysis method given in Ref 8 and is shown by the flow diagram in Fig 4.

Comparison with other data

The author has carried out several analyses of aligned 360° circumferential groove crankshaft bearings, for comparison with experimental and theoretical displacement and velocity coefficients derived by Parkins[2]. This has involved computing the usual eight displacement and velocity coefficients (relevant to bearings not subject to steady or dynamic misalignment) for both 'zero' amplitude and a range of finite amplitudes. Parkins' theoretical work incorporated a finite difference solution of Reynolds equation with the oil film viscosity varying as a function of temperature and pressure. Good agreement with the author's results was obtained[1].

Parkins[7] has also presented both experimental and theoretical data for the steady load performance of 360° circumferential groove bearings, covering both variable viscosity and isoviscous assumptions. Eight cases have been run for comparison with Parkins' isoviscous theoretical data, and the results presented in Table 1. It should be noted that Parkins adjusted his values of viscosity and cavitation pressure to obtain agreement between measured and theoretical load and attitude angle. In this paper the same values of viscosity and cavitation pressure have been used as input data.

The predicted extent of the cavitation zone for a 360° circumferential groove bearing is shown in Fig 5. This illustrates the effects of supply pressure variation, and includes comparitive theoretical data obtained by private communication with Mr F.A. Martin of the Glacier Metal Co Ltd. The Glacier results were obtained using a 72 circumferential x 15 finite difference mesh for each half of the bearing, while the present author used 72 x 14 elements.

For steadily loaded misaligned sterntube bearings, four

Table 1 Comparison of crankshaft bearing results with data in Ref 7

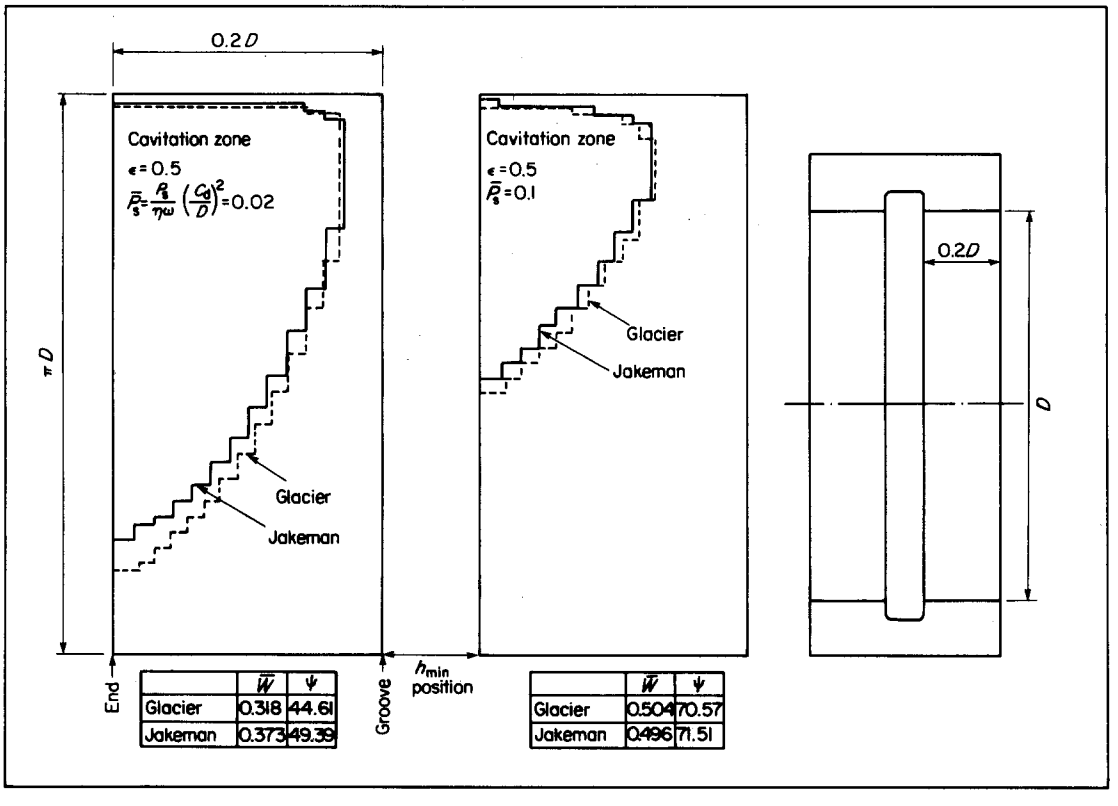
Case	Common input data					Parkins*		Jakeman	
	ϵ	C_d, mm	r/min	η, N s/m^2	P_c, N/mm^2 (gauge)	W, N	ψ°	W, N	ψ°
1	0.790	0.0909	1180	0.04470	-0.0069	670.2	38.70	683.2	39.28
2	0.864	0.0941	1180	0.04139	-0.0965	1341.2	30.60	1352.0	31.59
3	0.869	0.0900	2200	0.01883	-0.1719	1285.2	31.84	1323.9	33.16
4	0.902	0.0952	1500	0.02897	-0.0896	2219.2	24.00	2212.0	24.84
5	0.917	0.0936	2900	0.01069	-0.2136	2230.3	23.30	2182.2	24.49
6	0.926	0.0968	1500	0.02414	-0.0413	3120.3	19.60	3044.6	20.29
7	0.930	0.0983	2200	0.01552	0	3125.0	18.70	3091.5	19.15
8	0.942	0.1003	2900	0.008794	-0.1171	3113.2	18.00	3089.6	18.63

Bearing data: Diameter - 63.5 mm; overall length = 23.68 mm; groove width = 5.08 mm (360° circumferential in bearing axial centre); supply pressure = 0.2067 N/mm^2

**Table 10 of Ref 7 calculated data*

cases have been computed for comparison with theoretical data presented by Pinkus and Bupara[9]. These results are given in Table 2 together with results given by Martin in the discussion of the above paper. It should be noted that all moments are positive for consistency with the author's sign convention. A further misaligned sterntube bearing case has been produced for comparison with the theoretical example given by Hill and Martin[10]. These data are shown in Table 3.

The above details show the theoretical results presented here to be in good agreement with published and other data derived by finite difference solutions of Reynolds equation.



	$\bar{W}$	ψ
Glacier	0.318	44.61
Jakeman	0.373	49.39

	$\bar{W}$	ψ
Glacier	0.504	70.57
Jakeman	0.496	71.51

Fig 5 Comparison of predicted cavitation zones

Table 2 Comparison of sterntube bearing results with examples in Ref 9

		Case 1	Case 2	Case 3	Case 4
Load vector angle	Pinkus & Bupara	3°	1°	0°	2°
	Jakeman	3.4°	0.9°	0.3°	1.8°
Dimensionless parameters					
Eccentricity ratio	All	0.4	0.4	0.8	0.8
Load	Pinkus & Bupara	3.56	4.13	22.3	26.5
	Martin	3.56	4.13	22.3	26.5
	Jakeman	3.60	4.29	22.5	27.35
Vertical moment	Pinkus & Bupara	0.26	0.55	1.67	3.56
	Martin	0.27	0.58	1.5	3.2
	Jakeman	0.261	0.589	1.698	3.826
Minimum film thickness	Pinkus & Bupara	0.33	0.14	0.10	0.04
	Martin	0.34	0.15	0.105	0.05
	Jakeman	0.334	0.137	0.10	0.041
Vertical journal slope	Pinkus & Bupara	0.37	0.59	0.10	0.04
	Martin	0.37	0.59	0.10	0.16
	Jakeman	0.369	0.591	0.112	0.179

Bearing with 2 axial grooves subtending 30° are each at 90° and 270° from top; L/D = 1

Table 3 Comparison of sterntube bearing results with example in Ref 10

	Hill and Martin	Jakeman
Misalignment (vertical), rad	0.0002	0.0002
Min. film thickness, mm	0.123	0.123
Load, N	768 000	770 150
Moment, N mm	6.451×10^7	6.591×10^7

Bearing data: 800 mm diameter × 1200 mm long; 1.4 mm diametral clearance; 80 r/min. Axial grooves at ± 90° from top; film viscosity 0.125 N s/m²

Further development

Where substantial journal amplitudes are involved, the adequacy of displacement and velocity coefficients is doubtful in view of their nonlinearity. For such conditions a journal orbit analysis is more appropriate. The numerical analysis method described in this paper can be readily adapted for use in this type of analysis. It is considered important, however, that the oil film history concept[6] should be incorporated. Since the author's approach is based directly upon continuity considerations, application of oil film history should be quite straight forward.

Considering oil flow through the cavitation zone, the oil flow will be purely circumferential since it is entirely due to journal surface velocity (with the exception of axial element boundaries at pressures differing from the cavitation pressure). It follows then, that in a steady state condition, the circumferential oilflow rate through a column of elements is constant throughout the cavitation zone. The flow rate through the cavitation zone is therefore mainly dictated by the film thickness at the upstream boundary. In a dynamic situation the film thickness at the upstream cavitation zone boundary will vary, thus the flow rate through the cavitation zone will be subject to a corresponding variation. Consider a small quantity of oil which passes the upstream cavitation zone boundary at axial position I and time $t1$ and subsequently passes element J, I at time $t2$. The flow rate through element J, I at time $t2$ is thus determined by the film thickness at the upstream cavitation zone boundary at axial position I and time $t1$.

No work involving variable viscosity oil films or elastohydrodynamic situations has been carried out and no special problems are envisaged in applying the analysis technique described in this paper to such areas. For any particular application, the justification for the substantial additional complexity of variable viscosity and elastohydrodynamics should be carefully weighed.

Conclusions

This paper presents a numerical analysis method which is essentially simple yet has demonstrated a perfectly acceptable standard of accuracy for practical purposes. More sophisticated analytical techniques undoubtedly exist, but the simplicity of this method should enable a much clearer understanding to be attained of the way in which the physical processes are being modelled.

It is the author's view that the most difficult part of any analytical work is in conceiving the nature of the physical processes involved. Developing a mathematical model for these processes is, or should be, the simpler part of the problem. The method advocated in this paper should enhance the 'feel' for the physical processes by avoiding undue mathematical complexity.

Appendix

Derivation of equations

Film pressures: For laminar flow conditions and assuming that the rectangular film elements are sufficiently small to consider the bearing and journal surfaces as parallel flat plates, the flow rates across the element boundaries may be expressed as:

Circumferential direction

$$h\,\Delta a\,\frac{U}{2} - h\,\Delta a\,\frac{h^2}{12\eta}\,\frac{\partial P}{\partial c}$$

Axial direction

$$-h\,\Delta c\,\frac{h^2}{12\eta}\,\frac{\partial P}{\partial a}$$

Assuming linear pressure gradients between the centre of the element under consideration and the centres of the axially and circumferentially adjacent elements, the flow rates across the four element boundaries may be written thus:

$$Q_{ci} = (h_a + h_b)(U + V_T)\,\Delta a/4 - H_{ci}(P - P_{ci})$$

$$Q_{ai} = -H_{ai}(P - P_{ai})$$

$$Q_{co} = (h_c + h_d)(U + V_T)\,\Delta a/4 - H_{co}(P_{co} - P)$$

$$Q_{ao} = -H_{ao}(P_{ao} - P)$$

Note that the tangential component of the journal axis velocity V_T has been added to the journal surface velocity due to rotation U. In addition, the viscous flow functions H_{ci}, etc., have been incorporated, these being based upon the mean film thickness for the boundaries concerned.

The continuity equation assuming the lubricant to be incompressible is:

$$Q_{ci} + Q_{ai} + V_N\,\Delta c\,\Delta a - Q_{vi} = Q_{co} + Q_{ao} - Q_{vo}$$

The component of the journal axis velocity normal to the surface at the element centre V_N is assumed to be the effective mean value for the element. The terms Q_{vi} and Q_{vo} represent the circumferential flow rates of gas or vapour across the upstream and downstream element boundaries respectively, the density of which is assumed to be negligible relative to the liquid phase of the lubricant.

Substituting the expressions for Q_{ci} etc., into the continuity equation we have:

$$(h_a + h_b)(U + V_T)\,\Delta a/4 - H_{ci}(P - P_{ci}) - H_{ai}(P - P_{ai})$$
$$+ V_N\,\Delta c\,\Delta a - Q_{vi} = (h_c + h_d)(U + V_T)\,\Delta a/4$$
$$- H_{co}(P_{co} - P) - H_{ao}(P_{ao} - P) - Q_{vo}$$

Substituting for the nett velocity-induced flow rate K, this reduces to:

$$- H_{ci}(P - P_{ci}) - H_{ai}(P - P_{ai}) - Q_{vi}$$
$$= K - H_{co}(P_{co} - P) - H_{ao}(P_{ao} - P) - Q_{vo}$$

During the film pressure relaxation process it is initially assumed that for each element $P > P_c$, and consequently that $Q_{vo} = 0$.

One may therefore solve for P by re-arranging this equation to:

$$P = \frac{H_{ci}P_{ci} + H_{ai}P_{ai} + H_{co}P_{co} + H_{ao}P_{ao} - K - Q_{vi}}{H_{ci} + H_{ai} + H_{co} + H_{ao}}$$

If the above equation results in $P < P_c$ then P is made equal to P_c and the continuity equation now becomes:

$$-H_{ci}(P_c - P_{ci}) - H_{ai}(P_c - P_{ai}) - Q_{vi}$$
$$= K - H_{co}(P_{co} - P_c) - H_{ao}(P_{ao} - P_c) - Q_{vo}$$

Thus a finite Q_{vo} may be determined to satisfy continuity with $P = P_c$ from:

$$Q_{vo} = K + Q_{vi} - (H_{ci}P_{ci} + H_{ai}P_{ai} + H_{co}P_{co} + H_{ao}P_{ao}) + (H_{ci} + H_{ai} + H_{co} + H_{ao})P_c$$

Power loss: To calculate power loss it is necessary to determine the tangential force exerted by each element on the journal surface in the circumferential direction. The viscous shear stresses from which the above force may be derived are: $\eta U/h$ due to surface velocity and $(h/2)\,\partial P/\partial c$ due to the pressure gradient. With the element configuration adopted, these viscous shear stresses may only be calculated at the element boundaries. The viscous shear stress for the complete element is therefore taken as the mean value for the two circumferential boundaries:

$$\tau = \frac{1}{2}\left[\frac{2\eta U}{h_a + h_b} + \frac{2\eta U}{h_c + h_d}\right] + \frac{1}{2}\left[\frac{1}{2}\left(\frac{h_a + h_b}{2}\right)\left(\frac{P - P_{ci}}{\Delta c}\right) + \frac{1}{2}\left(\frac{h_c + h_d}{2}\right)\left(\frac{P_{co} - P}{\Delta c}\right)\right]$$

$$\therefore\ \tau = \eta U\left[\frac{1}{h_a + h_b} + \frac{1}{h_c + h_d}\right] + \left[\frac{(h_a + h_b)(P - P_{ci}) + (h_c + h_d)(P_{co} - P)}{8\,\Delta c}\right]$$

Where a full film exists:

$$F_c = \tau\,\Delta a\,\Delta c$$

Where cavitation exists, however, it is assumed that the lubricant and gas/vapour form discrete streams of rectangular cross section. Δa is thus effectively reduced to B_{ai} and B_{ao} at the upstream and downstream circumferential boundaries respectively, where:

$$B_{ai} = \Delta a - \frac{4Q_{vi}}{U(h_a + h_b)}$$

$$B_{ao} = \Delta a - \frac{4Q_{vo}}{U(h_c + h_d)}$$

Since the pressure gradient terms in the equation for τ automatically became zero in the cavitation zone, no modification of Δa with respect to these terms is necessary, therefore we may write:

$$F_c = \eta U \Delta c\left[\frac{B_{ai}}{h_a + h_b} + \frac{B_{ao}}{h_c + h_d}\right] + \left[(h_a + h_b)(P - P_{ci}) + (h_c + h_d)(P_{co} - P)\right]\Delta a/8$$

Power loss may then be calculated by summating the above tangential forces and multiplying by the journal surface velocity:

$$H = U\sum_{J=1,\,I=1}^{J=M_c,\,I=N_a} F_c$$

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